MCQ A · definitions and operations

After S16. MCQ B after S20; fixed point after S31.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. $u_n = 3 + \frac{2}{n+1}$. Its limit is:

A. 0 B. 2

C. 3 D. $+\infty$

2. $\frac{2n^2+1}{5n^2+n}$ tends to:

A. $\frac{2}{5}$ B. $\frac{5}{2}$

C. 0 D. $+\infty$

3. $u_n \to 2$ and $v_n \to 0$. What can be stated without any further assumption?

A. $u_n / v_n \to +\infty$ B. $u_n v_n \to 0$

C. $u_n / v_n \to 0$ D. $u_n + v_n \to 0$

4. For $u_n = 2 + \frac{3}{n+1}$, the first index from which $|u_n - 2| < 0.1$ is:

A. 28 B. 29

C. 31 D. 30

5. Sequence $(-1)^n$ is:

A. converging to 0 B. bounded and divergent

C. increasing D. diverging to $+\infty$

MCQ B · choosing a theorem

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. We know that u_n is increasing and $u_n \leq 5$. Necessarily:

A. $u_n \to 5$ B. $u_n \geq 0$

C. $u_n \to \ell \leq 5$ D. u_n constant

7. $u_n = \frac{\sin n}{n}$ for $n \geq 1$. The direct tool is:

A. monotone convergence

B. a fixed-point equation

C. the law of large numbers

D. the squeeze theorem

8. An increasing sequence that is not bounded above:

A. tends to $+\infty$

B. may tend to 0

C. is always geometric

D. oscillates between two values

9. $\sqrt{n^2+n} - n$ tends to:

A. $+\infty$

B. 1

C. $\frac{1}{2}$

D. 0

10. If $0 \leq u_n \leq (\frac{3}{4})^n$ for every index:

A. u_n is geometric

B. $u_n \to 0$

C. u_n is decreasing

D. $u_n \to \frac{3}{4}$

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. Every convergent sequence is monotonic.

2. Every increasing bounded sequence converges.

3. If $u_n \to \ell$ and $u_n < 5$, then $\ell < 5$.